

Statistical Distribution and Probability Analysis for E-commerce Transactions

This project analyzes customer transaction behavior using statistical distributions, probability concepts, transformations, and visualization techniques. The project focuses on understanding transaction patterns, identifying skewness, modeling probability distributions, and interpreting customer purchasing behavior using Python.

1. Bernoulli Distribution

Bernoulli Distribution was used to model transaction success and failure patterns. The dataset contains transaction status values such as Success and Fail. Since Bernoulli distribution works with binary outcomes, it is suitable for analyzing whether transactions succeed or fail.

Formula: $P(X=x)=p^x(1-p)^{(1-x)}$

Interpretation: Interpretation: The analysis showed that most transactions were successful, indicating stable transaction processing within the e-commerce platform.

2. Binomial Distribution

Binomial Distribution was used to calculate the probability of obtaining multiple successful transactions over a fixed number of transaction attempts.

Formula: $P(X=x)=nC_x p^x (1-p)^{(n-x)}$

Interpretation: Interpretation: The graph showed the probability of different numbers of successful transactions. The highest probability occurred near the average number of successful transactions.

3. Poisson Distribution

Poisson Distribution was used to model the number of customer transactions occurring within a fixed time interval. The average transaction count was used as lambda (λ).

Formula: $P(X=x)=e^{-\lambda}\lambda^x/x!$

Interpretation: Interpretation: The Poisson graph showed that moderate transaction counts occur more frequently while very high transaction counts are rare.

4. Log-Normal Distribution

Transaction amounts were positively skewed, meaning most customers made smaller purchases while few customers made extremely large purchases. Therefore, Log-Normal Distribution was used to model transaction amount behavior.

Formula: $Y = \ln(X)$

Interpretation: Interpretation: After logarithmic transformation, the distribution became more symmetrical, indicating that the transaction data follows a log-normal pattern.

5. Power Law Distribution

Power Law Distribution was used to analyze heavy-tailed transaction behavior where a small number of customers contributed very large transaction amounts.

Formula: $P(x) = Cx^{-\alpha}$

Interpretation: Interpretation: The Power Law graph showed long-tail behavior, confirming that few customers generated high-value transactions.

6. Q-Q Plot

Q-Q Plot was generated to compare transaction data with a normal distribution. This plot helps determine whether the data follows normality.

Formula: $Q_{\text{sample}} = Q_{\text{normal}}$

Interpretation: Interpretation: The points deviated from the straight line, indicating skewness and non-normal distribution in the original transaction data.

7. Box-Cox Transformation

Box-Cox Transformation was applied to reduce skewness and stabilize variance. This transformation improves the distribution shape for statistical analysis.

Formula: $Y(\lambda) = (X^\lambda - 1)/\lambda$

Interpretation: Interpretation: After transformation, the distribution became more balanced and closer to normal distribution.

8. Z-score Analysis

Z-score analysis was used to identify outlier transactions and measure how far transaction values deviated from the mean.

Formula: $Z = (X - \mu)/\sigma$

Interpretation: Interpretation: Transactions with high Z-score values represented unusual or extremely large customer purchases.

9. PDF and CDF Analysis

PDF and CDF plots were generated to analyze transaction probability density and cumulative transaction probability behavior.

Formula: PDF= $f(x)$, CDF= $P(X \leq x)$

Interpretation: Interpretation: PDF showed where transaction amounts were concentrated, while CDF showed cumulative transaction probability as transaction amount increased.

Graphs Used in the Project

Graph	Purpose
Histogram	Shows transaction amount distribution
Bernoulli Bar Chart	Compares success and fail probability
Binomial Distribution Graph	Shows successful transaction probabilities
Poisson Distribution Plot	Models transaction frequency
Q-Q Plot	Checks data normality
Box-Cox Histogram	Shows transformed data distribution
Power Law Curve	Visualizes long-tail transaction behavior
PDF Plot	Shows probability density
CDF Plot	Shows cumulative probability
Z-score Plot	Identifies outliers

Final Conclusion

The project successfully applied statistical distributions, probability analysis, and visualization techniques to understand customer transaction behavior. The analysis showed that transaction amounts are positively skewed and best fit Log-Normal and Power Law distributions. Poisson distribution effectively modeled transaction frequency, while Box-Cox transformation improved distribution normality. Overall, the project demonstrated how statistical techniques help businesses understand purchasing patterns, detect unusual transactions, and support better decision-making in e-commerce systems.